

Markscheme

Mathematics: analysis and approaches

Practice question bank

143. (a)

Points A , B and C on an Argand diagram represent $z_1 = 1 + i$, $z_2 = 4 + i$, and $z_3 = 4 + 5i$.

Find the area of triangle ABC .

$$AB = |z_2 - z_1| = 3 \text{ (horizontal)}, BC = |z_3 - z_2| = 4 \text{ (vertical)} \quad (M1) \text{ A1}$$

since AB is horizontal and BC is vertical, angle $B = 90^\circ$ **R1**

$$\text{area} = \frac{1}{2}(3)(4) \quad (M1)$$

$$= 6 \quad \text{A1}$$

[5 marks]